

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA0211

COURSE NAME: C Programming

COURSE OUTCOME COVERED

CO1: Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)

Assessment Tool Weightage: 10%

QUESTIONS:

Total Marks: 100

Annexure A

MCQ

Code of Conduct:

I B. Ganga Jayadev Reddy 192472248 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

ASSESSMENT TOOL 1

CO1: Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)

Q1. A programmer wants to display the octal equivalent of a decimal number stored as 25 using printf. Which output will the program produce?

```
int n = 25;  
printf("%o", n);
```

Output?

- A) 31
- B) 25
- C) 32
- D) Error

Q2. A developer prints an integer value 26 using the %x format specifier. Considering lowercase hexadecimal representation in C, what will be the output?

```
int n = 26;  
printf("%x", n);
```

Output?

- A) 1A
- B) 1a
- C) 26
- D) Error

Q3. Nerd initializes a variable using 017 and prints it using %d. Since the value is stored in a different number system, what will be the resulting output?

```
int n = 017;  
printf("%d", n);
```

Output?

- A) 17
- B) 15
- C) 16
- D) Error

Q4. A variable is assigned a hexadecimal constant 0xA5 and printed using %d. After decimal conversion, what value will be displayed?

```
int n = 0xA5;  
printf("%d", n);
```

Output?

- A) 155
- B) 165
- C) 150
- D) Error

Q5. Geek uses 010 as the value of an integer and prints it using %x. After converting the value appropriately, what will be the final output?

```
int n = 010;  
printf("%x", n);  
Output?  
A) 8  
B) A  
C) 10  
D) Error
```

Q6. A technician adds a hexadecimal value 0x10 and an octal value 010, then prints the result using %d. What will be the output?

```
printf("%d", 0x10 + 010);  
Output?  
A) 24  
B) 18  
C) 20  
D) 16
```

Q7. A variable with value 20 is printed using both %o and %x format specifiers in the same statement. What will be the output?

```
int n = 20;  
printf("%o %x", n, n);  
Output?  
A) 24 14  
B) 20 20  
C) 24 20  
D) Error
```

Q8. A macro #define A 10+5 is used in an expression A*2 without parentheses. What will be the output?

```
#define A 10+5  
printf("%d", A*2);  
Output?  
A) 30  
B) 25  
C) 40  
D) Error
```

Q9. A macro #define X 10 is defined, and later a variable with the same name int X = 5; is declared. What will happen when printing X?

```
#define X 10  
int X = 5;  
printf("%d", X);  
Output?  
A) 10  
B) 5  
C) Error  
D) Garbage
```

Q10. A constant variable is declared as "const int x = 10" and then reassigned using "x = x + 5". What will be the result?

```
const int x = 10;
```

```
x = x + 5;
```

Result?

A) 15

B) 10

C) Error

D) Garbage

Q11. A student divides an integer (5) by a float (2) and prints the result using %f. What will be the output?

```
int a = 5;
```

```
float b = 2;
```

```
printf("%f", a/b);
```

Output?

A) 2.000000

B) 2.500000

C) 2

D) Error

Q12. A float variable is initialized as float x = 5/2 and then printed using %f. What will be the output?

```
float x = 5/2;
```

```
printf("%f", x);
```

Output?

A) 2.500000

B) 2.000000

C) 5

D) Error

Q13. A character variable is assigned the value 100 and printed using %c. What will be displayed?

```
char c = 100;
```

```
printf("%c", c);
```

Output?

A) 100

B) d

C) Error

D) Garbage

Q14. A macro, A, is defined and then undefined using #undef. What will happen when attempting to print A?

```
#define A 5
```

```
#undef A
```

```
printf("%d", A);
```

Output?

A) 5

B) 0

- C) Error
- D) Garbage

Q15. A variable is declared but not initialized, and then printed. What will be the output?

```
int a;  
printf("%d", a);  
Output?  
A) 0  
B) Error  
C) Garbage value  
D) 1
```

Q16. A character, 'A', is stored in a variable, and 1 is added to it before printing the result as an integer. What will be the output?

```
char a = 'A';  
int b = a + 1;  
printf("%d", b);  
Output?  
A) 66  
B) 65  
C) A  
D) Error
```

Q17. A macro `#define B 2+3` is used in the expression `B + 4`. What will be the output?

```
#define B 2+3  
printf("%d", B+4);  
Output?  
A) 9  
B) 10  
C) 11  
D) Error
```

Q18. A macro `#define X 3+2` is used in the expression `X*X` without parentheses. What will be the output?

```
#define X 3+2  
printf("%d", X*X);  
Output?  
A) 25  
B) 11  
C) 19  
D) Error
```

Q19. An integer is divided by a float and printed using `%f`. What will be the output of `10/3`?

```
int a = 10;  
float b = 3;  
printf("%f", a/b);  
What will be printed?  
A) 3.333333  
B) 3.000000
```


- C) 3
D) Error

Q20. A character 'A' is printed using %d. What will be the output?

```
char ch = 'A';
printf("%d", ch);
```

Output?

- A) A
B) 65
C) Error
D) Garbage

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. MCQ'S

1. `int n=25;`
`printf("%o", n);`
output = 31 (A)
2. `int n=26;`
`printf("%x", n);`
output = 1a (B)
3. `int n=17`
`printf("%d", n);`
output = 15 (B)
4. `int n = 0xA5`
`printf("%d", n);`
output = 165 (B)
5. `int n = 010;`
`printf("%x", n);`
output = 8 (A)
6. `printf("%d", 0x10+010)`
output = 24 (A)
7. `int n=20`
`printf("%o %x", n, n);`
output = 24 14 (A)
8. `#define A 10+5`
`printf("%d", A*2);`
output = 25 (B)
9. `#define X 10`
`int X=5`
`printf("%d", X);`
output = Error (C)
10. `const int X=10;`
`X=X+5;`
Result = Error (C)